Regression Analysis Project Report

Title

Impact of Advertising Spend, Salespersons, Customer Visits, and Discount Percentage on Sales Revenue: A Regression Analysis Approach

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1. Objective

The objective of this project was to identify the key factors influencing Sales Revenue using regression analysis techniques. The study examined how Advertising Spend, Salespersons, Customer Visits, and Discount Percentage affect Sales Revenue and determined the most significant predictors.

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2. Dataset Description

The dataset contains **105 observations** with one dependent variable and four independent variables.

Dependent Variable:

- Sales Revenue

Independent Variables:

- Advertising Spend
- Salespersons
- Customer Visits
- Discount %

Microsoft Excel was used to perform all statistical analyses and visualization.

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3. Methodology

The following analytical techniques were applied:

- Exploratory Data Analysis (EDA)
- Correlation Analysis

- Simple Linear Regression
- Multiple Linear Regression
- Multicollinearity Analysis (Correlation Matrix and VIF)
- Model Selection
- Residual Analysis

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4. Findings and Results

Exploratory Data Analysis (EDA)

The descriptive statistics indicated sufficient variation across all variables, confirming that the dataset was appropriate for regression analysis.

Correlation Analysis

Correlation analysis showed that **Advertising Spend** had the strongest positive relationship with Sales Revenue (**r = 0.718**). Salespersons (**r = 0.376**) and Customer Visits (**r = 0.206**) also showed positive relationships, while Discount Percentage (**r = 0.071**) had a very weak relationship with Sales Revenue.

Simple Regression Analysis

A simple regression model was developed using Advertising Spend as the predictor.

Regression Equation:

Sales Revenue = 48786.13 + 2.484(Advertising Spend)

The model achieved an **R^2 value of 0.516**, indicating that Advertising Spend alone explained approximately **51.6%** of the variation in Sales Revenue.

Multiple Regression Analysis

A multiple regression model was developed using all four independent variables.

Regression Equation:

Sales Revenue = 13258.47 + 2.501(Advertising Spend) + 1953.01(Salespersons) + 5.501(Customer Visits) + 255.03(Discount %)

The model achieved an **R^2 value of 0.714**, indicating that the independent variables collectively explained approximately **71.4%** of the variation in Sales Revenue.

Significance Testing

Variable	P-value	Result
Advertising Spend	2.57E-24	Significant
Salespersons	4.73E-11	Significant
Customer Visits	0.000351	Significant
Discount %	0.1347	Not Significant

The results show that Advertising Spend, Salespersons, and Customer Visits significantly affect Sales Revenue, while Discount Percentage does not have a statistically significant impact.

Multicollinearity Analysis

Correlation values among independent variables were very low and VIF values were approximately equal to 1. Therefore, no multicollinearity problem was detected.

Model Selection

Using the Backward Elimination method, Discount Percentage was removed because its p-value was greater than 0.05.

Final Selected Model:

Sales Revenue = 13258.47 + 2.501(Advertising Spend) + 1953.01(Salespersons) + 5.501(Customer Visits)

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5. Conclusion

The analysis demonstrated that **Advertising Spend, Salespersons, and Customer Visits are the most important factors influencing Sales Revenue**. The final regression model explained **71.4%** of Sales Revenue variation and showed strong statistical significance.

Based on the findings, organizations can improve Sales Revenue by increasing advertising investment, strengthening sales teams, and enhancing customer engagement activities. The final model provides a reliable framework for understanding and predicting Sales Revenue performance.